

SEMESTER PROJECT	Report Title	Database Schema
	Course	CMPS3162 Advanced Databases
	Topic	Hotel Room Booking & Housekeeping Management
	Due Date	Feb 19, 2026

DATABASE SCHEMA

CMPS3162 Advanced Databases (Quiz 2)
Hotel Room Booking & Housekeeping Management
Version 1.0.0
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Declaration of Use for Generative AI in Assessments



I hereby declare that in the planning, drafting, and/or revision of the work attached, I have made use of Generative AI tools in the following ways:

Acknowledgement of Generative AI Tools Used

- | | |
|---|--|
| ☒ For brainstorming, | ☒ To generate translations of primary/secondary source content for consultation. |
| ☐ To find sources. | ☐ To generate translations included in the submitted work, whether or not manually revised. |
| ☐ To plan the structure/outline of the work. | ☐ To improve the language of my own phrases, sentences, and/or paragraphs. |
| ☐ To generate programming code. | ☐ To generate the text of (part of) the submitted work. |

Acknowledgement of Assessment Submission

I, **Andres Hung**, hereby confirm that on **February 16, 2025**:

1. I am the author of this submitted document.
2. I am responsible for any AI-generated errors or fabrications.
3. I understand the limitations and risks of using AI.
4. I used AI tools ethically, protecting all sensitive information.
5. I ensure any AI-assisted work remains originally my own.
6. I have appropriately acknowledged all use of generative AI.
7. Undeclared AI use constitutes academic dishonesty, which I acknowledge.
8. I am accountable for any resulting academic misconduct.

Please add more rows to the table as needed so that each tool used in the creation of your assessment submission is included in the declaration.

Generative AI Tool Used (Please List Each Separately)	Purpose of Use	Briefly Explain the Extent of Use
ChatGPT (Free)	Getting suggestions for ERD and relational schema.	Usage involved providing the draw.io XML code to determine whether some relationships could be better modeled. Also asked whether some entity sets should be strong or weak.

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Course Code: CMPS3162

Signature: 

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Date: February 16, 2025

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Design Overview

This report outlines the database schema design for a simple hotel chain with multiple hotels. Primarily covered are guest reservations and housekeeping management. Three requirements are highlighted:

- The system shall track housekeeping tasks and their completion times.
- The system shall track maintenance issues. (It is assumed that maintenance issues concern hotel rooms and that only housekeepers manage them.)
- The system shall handle group bookings.

Three sections show the evolution of the design. The [Conceptual Database Design section](#) highlights the entity and relationship sets, using entity relationship diagram (ERD) notation from Korth et al. (2019). The [Relational Schema section](#) is derived from the ERD and is categorized by strong entities, weak entities, etc. Lastly, the [Physical Schema](#) section shows the physical tables along with their attributes' PostgreSQL data types and constraints. The resulting diagram uses the crow's foot notation.

A number of normalization decisions were made to conform to the project's highlighted requirements. To handle group bookings, a separate **registration** entity between the **room** and **reservation** entities allows for multiple rooms to be associated with a single reservation. Housekeeping tasks and maintenance reports are tracked in their own tables. The **housekeeper** entity was designed as a disjoint specialization of the **employee** entity, which in turn is an overlapping specialization of the **person** entity. This design helps to separate hotel employees and hotel guests.

Critical indexes are expected to be created primarily on the primary and foreign keys of every physical table to support efficient database query performance, which will likely involve numerous JOINS with the current design.

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Conceptual Database Design

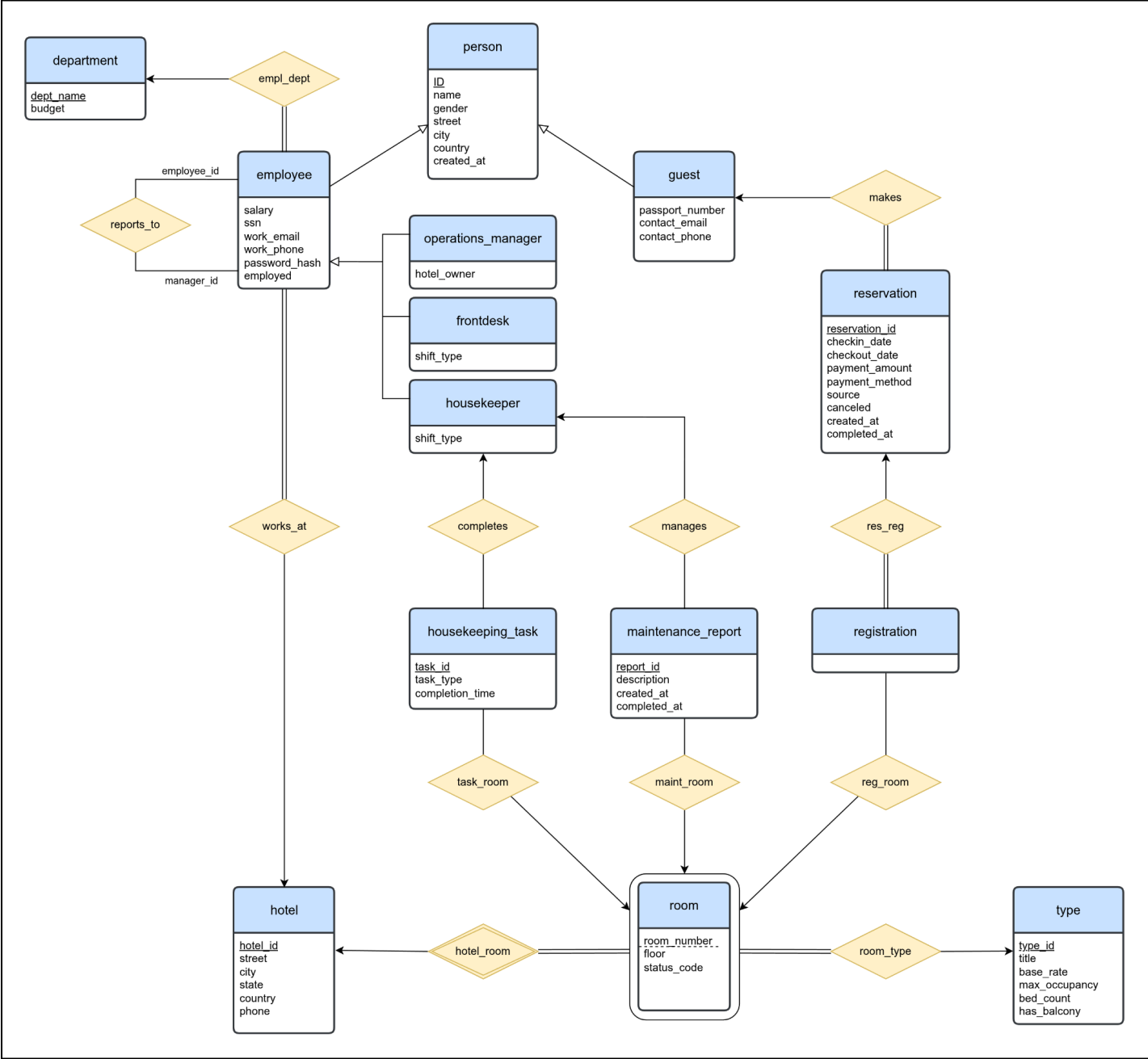


Figure 1 - Entity Relationship Diagram for the Hotel Room Booking & Housekeeping Management Database.

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Relational Schema

Strong Entity Sets

hotel (hotel_id, street, city, state, country, phone)

department (dept_name, budget)

type (type_id, title, base_rate, max_occupancy, bed_count, has_balcony)

reservation (reservation_id, checkin_date, checkout_date, payment_amount, payment_method, source, canceled, created_at, completed_at, guest_id)

registration (reservation_id, hotel_id, room_number)

housekeeping_task (task_id, task_type, completion_time)

maintenance_report (report_id, description, created_at, completed_at)

Weak Entity Sets

room (hotel_id, room_number, floor, status_code, type_id)

Relationship Sets

reports_to (employee_id, manager_id)

completes (ID, task_id)

manages (ID, report_id)

(Note: the rest of the relationship sets are redundant or combined)

Generalizations/Specializations

person (ID, name, gender, street, city, country, created_at)

guest (ID, passport_number, contact_email, contact_phone)

employee (ID, salary, ssn, work_email, work_phone, password_hash, employed, hotel_id, dept_name)

employee

operations_manager (ID, hotel_owner)

frontdesk (ID, shift_type)

housekeeper (ID, shift_type)

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Physical Schema

Tables

hotel	
PK	<u>hotel_id</u> BIGSERIAL
	street TEXT NOT NULL
	city TEXT NOT NULL
	state TEXT NOT NULL
	country TEXT NOT NULL
	phone TEXT NOT NULL

department	
PK	<u>dept_name</u> TEXT
	budget NUMERIC(18, 2)

type	
PK	<u>type_id</u> INT
	title TEXT NOT NULL
	base_rate NUMERIC(12, 2) NOT NULL
	max_occupancy INT NOT NULL
	bed_count INT NOT NULL
	has_balcony BOOL NOT NULL

reservation	
PK	<u>reservation_id</u> BIGSERIAL
	checkin_date TIMESTAMP(0) WITH TIME ZONE NOT NULL
	checkout_date TIMESTAMP(0) WITH TIME ZONE NOT NULL
	payment_amount NUMERIC(12, 2) NOT NULL
	payment_method TEXT NOT NULL
	source TEXT NOT NULL
	canceled BOOL NOT NULL
	created_at TIMESTAMP(0) WITH TIME ZONE NOT NULL DEFAULT NOW()
	completed_at TIMESTAMP(0) WITH TIME ZONE
FK	guest_id BIGINT NOT NULL REFERENCES guest.id

registration	
PK, FK	<u>reservation_id</u> BIGINT REFERENCES reservation.reservation_id ON DELETE CASCADE
PK, FK	<u>hotel_id</u> BIGINT REFERENCES room.hotel_id ON DELETE CASCADE
PK, FK	<u>room_number</u> BIGINT REFERENCES room.room_number ON DELETE CASCADE

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room	
PK, FK	<u>hotel_id</u> BIGINT REFERENCES hotel.hotel_id ON DELETE CASCADE
PK	<u>room_number</u> INT
	floor INT NOT NULL status_code TEXT NOT NULL CHECK (status IN ('clean', 'dirty', 'maintenance'))
FK	type INT NOT NULL REFERENCES type.type_id

housekeeping_task	
PK	<u>task_id</u> BIGSERIAL
	task_type TEXT NOT NULL completion_time INTERVAL NOT NULL
FK	housekeeper BIGINT NOT NULL
FK	hotel_id BIGINT NOT NULL REFERENCES room.hotel_id
FK	room_number BIGINT NOT NULL REFERENCES room.room_number

maintenance_report	
PK	<u>report_id</u> BIGSERIAL
	description TEXT NOT NULL created_at TIMESTAMP(0) WITH TIME ZONE NOT NULL DEFAULT NOW() completed_at TIMESTAMP(0) WITH TIME ZONE
FK	housekeeper BIGINT NOT NULL
FK	hotel_id BIGINT NOT NULL REFERENCES room.hotel_id
FK	room_number BIGINT NOT NULL REFERENCES room.room_number

person	
PK	<u>id</u> BIGSERIAL
	name TEXT NOT NULL gender TEXT street TEXT city TEXT country TEXT created_at TIMESTAMP(0) WITH TIME ZONE DEFAULT NOW()

guest	
PK, FK	<u>id</u> BIGINT REFERENCES person.id ON DELETE CASCADE
	passport_number TEXT UNIQUE NOT NULL contact_email CTEXT NOT NULL phone TEXT NOT NULL

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employee	
PK, FK	<u>id</u> BIGINT REFERENCES person.id ON DELETE CASCADE
	salary NUMERIC(18, 2) NOT NULL ssn TEXT UNIQUE NOT NULL work_email CITEXT UNIQUE NOT NULL work_phone CITEXT UNIQUE NOT NULL work_phone CITEXT UNIQUE NOT NULL password_hash BYTEA NOT NULL employed BOOL NOT NULL FK department TEXT NOT NULL REFERENCES department.dept_name FK hotel BIGINT NOT NULL REFERENCES hotel.hotel_id FK reports_to BIGINT NOT NULL REFERENCES person.id

operations_manager	
PK, FK	<u>id</u> BIGINT REFERENCES employee.id ON DELETE CASCADE
	hotel_owner BOOL NOT NULL

frontdesk	
PK, FK	<u>id</u> BIGINT REFERENCES employee.id ON DELETE CASCADE
	shift_type TEXT NOT NULL CHECK (shift_type IN ('day', 'night'))

housekeeper	
PK, FK	<u>id</u> BIGINT REFERENCES employee.id ON DELETE CASCADE
	shift_type TEXT NOT NULL CHECK (shift_type IN ('day', 'night'))

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Schema Diagram

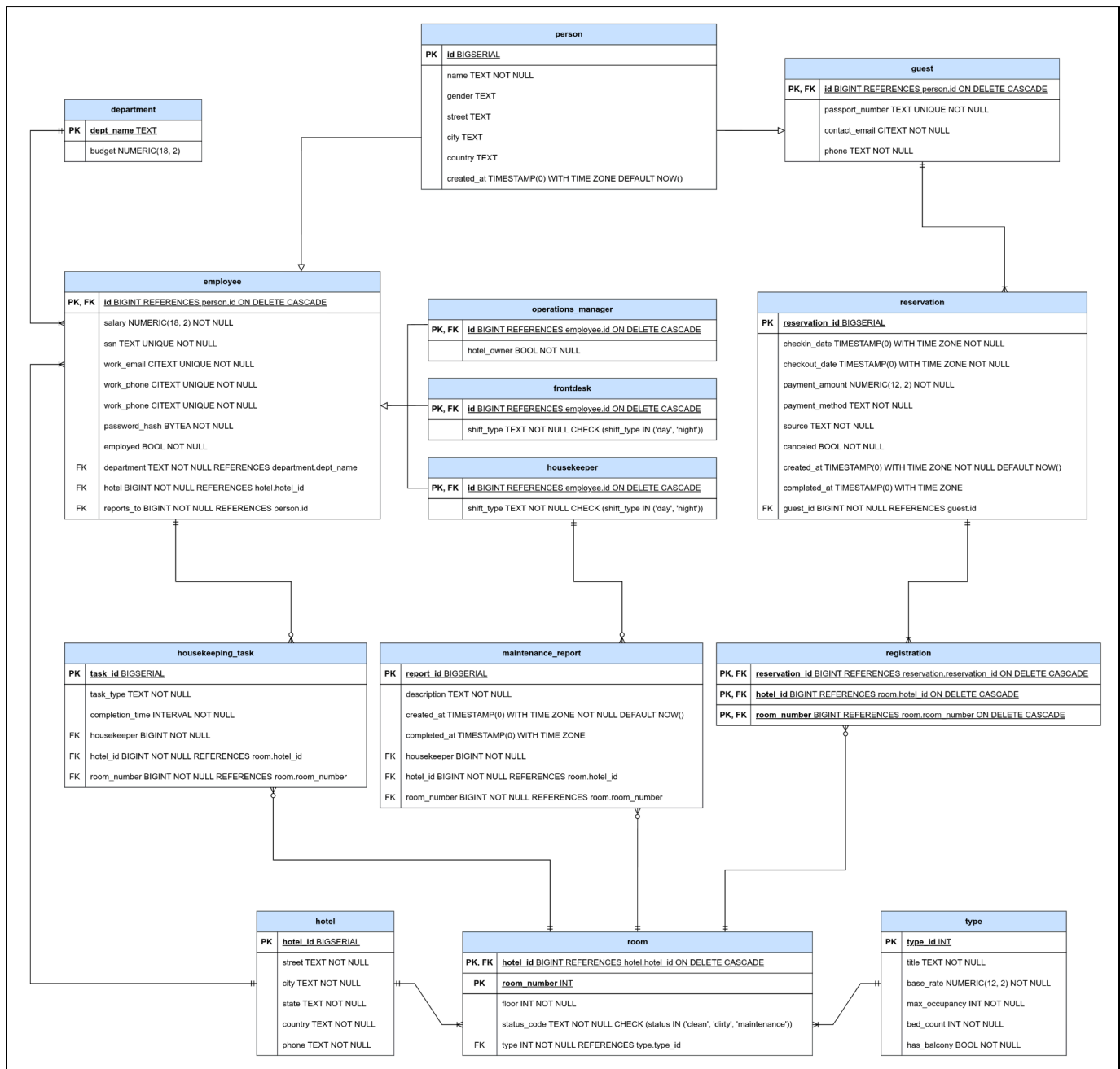


Figure 2 - Physical schema featuring primary keys, foreign keys, attributes, relationships, and PostgreSQL types.

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References

Korth, H. F., Sudarshan, S., & Professor, A. S. (2019). *Database System Concepts*. McGraw-Hill Education.